MODULATION & SPACE

BOB34

Five modular effects in one rack - a ladder low-pass, a ring modulator, a phaser, a resonant filter array and a saturator - in any order, each able to modulate the others.





Overview

BOB34 is a rack of five effects in the spirit of a classic American range of analogue effects pedals. The sound is designed from scratch. The signal runs through the modules in the order shown on THE CHAIN strip; the ◀ / ▶ keys on each module move it earlier or later, and the number beside them shows its place.

Every module has an IN key, its own DRIVE (an input stage that warms and crunches, level-compensated) and MIX. Every module also has a CV input: it can take another module's envelope, LFO or the noise generator, so one effect can move another - a phaser swept by the low-pass's envelope, a ring carrier pushed by the saturator, and so on.

Order changes run the new chain alongside the old one for 40 ms and then crossfade into it, and every key fades in 10 ms, so nothing clicks. The house output stage (CLIP, ceiling -0.3 dBFS) catches overs at the end.

Reach for it for

- Envelope wah and filter funk on guitar, bass, clavinet and synths
- Ring-mod bells, tremolo and metallic textures
- Slow and jet phasing
- Rhythmic filter patterns on pads and drums (MURF)
- Saturation that reacts to your playing, with noise for grit

---

Controls

The header

PRESETS	Opens the full preset list.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset name. Click it for the preset list too.
VFD / MATRIX	The panel's look: VFD charcoal and warm orange-brown, MATRIX battleship grey and wood cream. The sound does not change.
S / M / L	Panel size: small, medium or large.
?	The design credit, the GFX website and the address of this manual.

On every module

IN	The module in or out (a 10 ms fade).
◀ / ▶	Move the module earlier or later in the chain; the number shows its place (1 = first).
DRIVE	The module's input stage, 0 to 10: warmth, then crunch. Level-compensated. (On the SATURATOR, DRIVE is the effect itself.)
MIX	Dry / effect for the module, 0 to 100%.
CV	Click the CV key to pick the module's mod source - OFF, LP ENV, RING LFO, PH LFO, MURF ENV, SAT ENV or NOISE - and set CV AMT (-100 to +100%). It moves the LOWPASS's cutoff (up to 4 octaves), the RING's carrier (4 octaves), the PHASER's sweep (3 octaves), MURF's rate (2 octaves) or the SATURATOR's drive (12 dB).

LOWPASS

CUTOFF	A transistor-ladder low-pass, 20 Hz to 12 kHz.
RES	Resonance, 0 to 10: it sings, and self-oscillates near the top.
2-POLE / 4-POLE	The slope: 12 or 24 dB/oct (4-pole by default).
ENV	The envelope follower's amount, -10 to +10: up to 5 octaves of sweep, up with the level (positive) or down (negative).
FAST / SLOW	The envelope's response: snappy, or slow and smooth.

RING

CARRIER	A sine carrier, 0.6 Hz to 4 kHz: low settings are a tremolo, high ones bells and metal.
RATE / AMOUNT	An LFO sweeping the carrier: 0.1 to 25 Hz, up to 3 octaves.
SQUARE	The LFO's wave: sine, or square (lit) for stepped jumps.

PHASER

SWEEP	The notches' centre, 100 Hz to 8 kHz.
RES	Feedback: sharper, more vocal notches.
6 / 12-STAGE	The number of stages (lit = 12: more notches, a jet-like sweep).
RATE / AMOUNT	The LFO: 0.05 to 20 Hz, up to 2.5 octaves. MIX at 50% gives the deepest notches.

MURF

Faders 1 - 8	The levels of eight resonant band-pass filters, lowest on the left.
PATTERN	One of 12 built-in 16-step patterns (shown in the grid): each step fires some of the filters.
RATE	The pattern's speed, 0.5 to 20 steps per second.
ENVELOPE	Each step's shape, from short and percussive to long and swelling.
BASS / MID	The filters' voicing: lower (BASS) or higher (MID, lit).

SATURATOR

DRIVE	The drive, 0 to 10.
ENV	An envelope follower on the drive, -10 to +10: above 0 the drive rises with the level (harder, expanding), below 0 it falls (smoother, compressing).
NOISE / TONE	A noise generator mixed in, with its filter, 100 Hz to 12 kHz.
NOISE HP	The noise filter: low-pass, or high-pass (lit).

THE CHAIN and OUTPUT

THE CHAIN	Shows the order the modules run in; modules that are out are dimmed.
STEREO	Lit: the ring carrier's right side runs 90° ahead, the phaser's right LFO is inverted, MURF's filters split left / right and the noise is inverted on the right. Unlit: everything mono-linked.
LEVEL	The output, -12 to +12 dB, with the output meters beside it.
DRIFT	See DRIFT below (50% by default).

Knobs: drag up or down (Shift for fine moves) or use the mouse wheel; double-click to reset a knob to its default. A knob's value replaces its name while you touch it.

DRIFT

DRIFT is filter tuning and LFO rates: the ladder's and the phaser's frequencies wander a few cents, and the ring and phaser LFOs and MURF's clock wander a little. 50% is the standard amount, 100% twice that, 0% steady.

Presets

A preset sets the modules, the CV patch, the chain order and STEREO. OUTPUT, DRIFT and the look are left as they are. The list also has Reset all to defaults.

Init	The LOWPASS alone: 800 Hz, resonance 4, envelope +5, fast.
Funky Wah	LOWPASS, 2-pole, from 350 Hz with resonance 6.5 and envelope +8: an envelope wah.
Slow Sweep	PHASER into LOWPASS, the low-pass moved by the phaser's slow LFO, in stereo.
Ring Bells	RING at 640 Hz with a slow, gentle sweep, into an open LOWPASS; stereo.
Ring Tremolo	RING with a 6 Hz carrier: a stereo tremolo.
Jet Phaser	12-stage PHASER, resonance 8, a very slow deep sweep;

	stereo.
Classic Phase	6-stage PHASER, a moderate sweep, mono.
MuRF Pulse	MURF at six steps a second, short envelope; stereo.
MuRF Bass Steps	MURF on a falling fader slope, BASS voicing, longer envelope.
Saturated Wah	SATURATOR into LOWPASS, the wah also moved by the saturator's envelope.
Noise Wash	SATURATOR with filtered noise into LOWPASS and PHASER, the low-pass moved by the noise, the phaser sweeping slowly; stereo.
Full Rack	All five, SATURATOR first, the phaser moved by the low-pass's envelope; stereo.
Reset all to defaults	(Last in the list.) Init, with OUTPUT, DRIFT and the look back to their defaults.

Tips

- Order matters: SATURATOR before LOWPASS gives a fat, gritty wah; after it, the filter's resonance is driven into the saturation.
- For an auto-wah that follows the player, keep the LOWPASS on FAST and raise ENV; then patch LP ENV into the PHASER's CV.
- Try NOISE as a CV source at a small amount: everything it touches starts to shimmer.
- Switch STEREO off when the track has to stay solid in mono.

`File: gfx-bob34.jsfx`